

From the Logic of Existence to the Gauge Symmetry Group:

A Proof of Hypothesis $H_{\text{SU}2}$ from Axioms C1 and C2

Projective Dynamic Logo Framework — Document D60

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Abstract

Document D57 [1] derived $\sin^2 \theta_W^{(\text{tree})} = 1/4$ as an unconditional theorem of the PDL axioms C1–C4, and formulated Hypothesis $H_{\text{SU}2}$ as a well-motivated conjecture: that axiom C1 forces the Klein four-group V_4 to act trivially on the physical observable space of K_4 , making the effective symmetry group $G_{\text{eff}} = S_4/V_4 \cong S_3$. The present document proves this conjecture. The proof rests on two lemmas, each verified by independent exhaustive computation in Google Colab (scripts `PDL_C1V4_script1.py` [2] and `PDL_C1V4_script2.py` [3]) before any claim was incorporated into this document. A documented negative result is an integral part of the proof: the map $s \mapsto -s$ (global sign flip) does *not* preserve the coherence condition C2 on K_4 ; pulsation pairing is therefore not a naïve sign flip but the partition structure of the dynamic orbit $\mathcal{O}_4$ under V_4 . The proof proceeds as follows. The space $\text{Coh}(K_4)$ of coherent configurations decomposes into three S_4 -orbits of sizes $1 + 3 + 4$; the dynamic orbit $\mathcal{O}_4$ is isomorphic to V_4 as a regular V_4 -set. The three pulsation pairings of $\mathcal{O}_4$ coincide exactly with the three coset partitions of V_4 under its three order-2 subgroups H_1 , H_2 , H_3 . An element $g \in S_4$ preserves all three pairings if and only if the induced permutation σ_g of V_4 preserves all three coset structures — which holds if and only if $g \in V_4$ (Lemma B). Furthermore, V_4 fixes every element of $\mathcal{O}_1$ and $\mathcal{O}_3$ pointwise (Lemma C), so C1-admissibility is equivalent to V_4 -invariance on all of $\text{Coh}(K_4)$. The effective symmetry group is therefore $G_{\text{eff}} = S_4/V_4 \cong S_3$, an unconditional theorem of C1 and C2. Combined with D46 [4], D57 [1], D58 [5], and D59 [6], this completes the derivation of the Standard Model gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ and its fundamental representation as unconditional theorems of C1–C4, with no remaining conjectural step.

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1 Introduction and context

1.1 State of the programme at D59

The PDL programme derives physical laws and fundamental constants from four combinatorial axioms on finite signed graphs, with no presupposed spacetime, particles, or fields. The central object is the complete graph K_4 on four vertices and six edges, identified with the electron prototype [7]. The gauge sector has been built up across four documents. Document D46 [4] derived the $U(1)$ phase freedom of quantum mechanics as a structural consequence of the K_4 pulsation sign equivalence via the Hopf fibration, establishing Born’s rule at Level 2. Document D57 [1] derived the tree-level Weinberg angle $\sin^2 \theta_W^{(\text{tree})} = 1/4$ as an unconditional theorem via two independent routes (the AB-fraction route and the S_4 -orbit route), and identified V_4 as the group of pulsation pairs in the dynamic orbit $\mathcal{O}_4$. Document D58 [5] established $SU(3)$ as an unconditional algebraic theorem of C1–C4 via the Cartan–Killing–Weyl classification. Document D59 [6] identified the physical colour gauge group $SU(3)_c$ acting in the fundamental representation $\mathbf{3}$ on the carrier space $W = \{a + b + c = 0\} \subset \mathbb{C}^3$.

After D59, a single conjectural step remained in the gauge sector: Hypothesis H_{SU2} of D57, which stated that C1 forces V_4 to act trivially on the physical observable space, making $G_{\text{eff}} = S_4/V_4 \cong S_3$ an unconditional theorem rather than a well-motivated conjecture.

1.2 What D60 establishes

The present document proves H_{SU2} . The argument originates at the level of the logic of existence itself (Section 2): the co-originary status of the two states of a 2-cycle, which is constitutive of C1, forces every physical observable to be insensitive to the choice of representative within a pulsation pair. This is the definition of C1-admissibility. The remainder of the proof identifies, by algebraic means, the group that C1 and C2 together select as the symmetry group of this admissibility condition: it is V_4 , exactly, and for algebraic reasons that follow from the structure of K_4 alone.

The proof is entirely self-contained. All claims are either unconditional theorems of C1–C4 derived here, or established results of earlier corpus documents cited explicitly. No numerical approximation is introduced. Both verification scripts were executed independently in Google Colab before this document was drafted, in accordance with the PDL verrouillage protocol.

2 The logic of existence and axiom C1

2.1 Existence as repeatable distinction

The foundational question is not what things are, but what the minimum conditions are for anything to be at all. A distinction that cannot be recognised as such is indistinguishable from no distinction. For a distinction to be real — in the sense of being available to any description of the system — it must be repeatable: it must return. The minimum form of return is the 2-cycle: a configuration that is followed by a second, which is followed by the first again. This is the content of axiom C1.

Axiom 1 (C1 — Minimal binary pulsation, [8]). Existence is instantiated as a repeatable, non-trivial alternation between two discrete states, independent of any pre-existing temporal metric. The discrete dynamical system defined on the space of sign configurations must admit at least one logical 2-cycle: a configuration that returns to itself after exactly two update steps and is not a fixed point.

2.2 Co-originaryity and C1-admissibility

The decisive observation is the following. In a 2-cycle, the two states are not in an asymmetric relation: neither $s^{(1)}$ is prior to $s^{(2)}$, nor the reverse. They co-define one another. The system could equally well have been described with $s^{(2)}$ as the initial state; the resulting description would be physically identical.

This co-originary status is not a contingent feature of any particular 2-cycle: it is constitutive of the 2-cycle as such. If a description of the system attributed a different physical status to $s^{(1)}$ and $s^{(2)}$ — if some observable took different values on the two states — that observable would introduce a distinction that C1 itself does not contain. It would be an external input, not a consequence of the axiom.

Definition 2.1 (C1-admissible observable). A function $O : \text{Coh}(K_4) \rightarrow \mathbb{R}$ is *C1-admissible* if it is constant on every pulsation pair: $O(s^{(1)}) = O(s^{(2)})$ for every pulsation partner pair $\{s^{(1)}, s^{(2)}\}$ in the dynamic orbit $\mathcal{O}_4$.

Remark 2.2. Definition 2.1 is a direct formal expression of the co-originary principle: a C1-admissible observable is one that cannot distinguish the two states of a pulsation. Any observable that violates this condition introduces information that C1 does not provide, and is therefore not a consequence of the axiom. The definition is not an additional hypothesis; it is the logical content of C1 expressed at the level of the observable algebra.

3 Structure of $\text{Coh}(K_4)$ under S_4

3.1 Coherent configurations and the orbit decomposition

The coherence condition C2 requires that every triangle of K_4 carry a positive product of edge signs [7]. The set $\text{Coh}(K_4)$ of sign configurations satisfying C2 has exactly eight elements, organised into three S_4 -orbits.

Lemma: S_4 -orbit structure of $\text{Coh}(K_4)$

The automorphism group $S_4 = \text{Aut}(K_4)$ acts on $\text{Coh}(K_4)$ with three orbits:

$$\mathcal{O}_1 = \{(+, +, +, +, +, +)\}, \tag{1}$$

$$\mathcal{O}_3 = \{(-, -, +, +, -, -), (-, +, -, -, +, -), (+, -, -, -, -, +)\}, \tag{2}$$

$$\mathcal{O}_4 = \{(-, -, -, +, +, +), (-, +, +, -, -, +), (+, -, +, -, +, -), (+, +, -, +, -, -)\}, \tag{3}$$

where signs are listed in the order $(e_{01}, e_{02}, e_{03}, e_{12}, e_{13}, e_{23})$. This decomposition is verified exhaustively by script `PDL_C1V4_script1.py` (Section 2, 192 checks).

The orbit $\mathcal{O}_4$ is called the *dynamic orbit*: it is the orbit on which V_4 acts non-trivially, and it is the locus of the pulsation pairs. The orbits $\mathcal{O}_1$ and $\mathcal{O}_3$ are called *static* orbits.

3.2 Documented negative result: the sign flip does not preserve C2

The global sign flip $\tau : s \mapsto -s$ does *not* map $\text{Coh}(K_4)$ to itself. For every $s \in \text{Coh}(K_4)$, the configuration $-s$ is *not* coherent. Verified exhaustively: all 8/8 coherent configurations have their global flip outside $\text{Coh}(K_4)$.

Proof. A sign configuration s is coherent iff the product of signs on every triangle of K_4 equals $+1$. If $s \in \text{Coh}(K_4)$, then every triangle has product $+1$ under s . Under $-s$, every edge sign is negated; a triangle on three edges therefore has product $(-1)^3 = -1$. Hence $-s \notin \text{Coh}(K_4)$. $\square$

Remark 3.1. This negative result clarifies a potential ambiguity in the formulation of D46 [4], where the pulsation pair is described as $\{s^{(1)}, s^{(2)}\} = \{s, -s\}$. That description is valid in the amplitude space $\mathcal{H}_{\text{cyc}} \cong \mathbb{C}^2$, where $s^{(2)} = -s^{(1)}$ refers to the sign inversion of the amplitude basis vector, not of the edge sign configuration. At the level of $\text{Coh}(K_4)$, pulsation pairing is the partition structure of $\mathcal{O}_4$ induced by the action of V_4 .

4 Pulsation pairing structure and Lemma B

4.1 $\mathcal{O}_4$ as a regular V_4 -set

Lemma: Regular action of V_4 on $\mathcal{O}_4$

The group $V_4 = \{\text{id}, (01)(23), (02)(13), (03)(12)\}$ acts on $\mathcal{O}_4$ transitively and freely. In particular, $|\mathcal{O}_4| = |V_4| = 4$ and $\mathcal{O}_4 \cong V_4$ as a V_4 -set. Fixing the base point $s_0 = (-, -, -, +, +, +)$, the bijection

$$\varphi : V_4 \xrightarrow{\sim} \mathcal{O}_4, \quad v \mapsto v \cdot s_0 \tag{4}$$

is an explicit isomorphism of V_4 -sets.

Proof. Transitivity: from s_0 , the three non-trivial elements of V_4 produce the other three elements of $\mathcal{O}_4$ (verified exhaustively, script Section 3). Freeness: each non-trivial element of V_4 acts without fixed points on $\mathcal{O}_4$ (9 checks, all confirmed). $\square$

4.2 Pulsation pairings as coset partitions

Since $\mathcal{O}_4 \cong V_4$ as a V_4 -set, the structure of pulsation pairs is entirely determined by the group structure of V_4 . Let H_i denote the three order-2 subgroups of V_4 :

$$H_1 = \{e, (01)(23)\}, \quad H_2 = \{e, (02)(13)\}, \quad H_3 = \{e, (03)(12)\}. \quad (5)$$

Each H_i partitions V_4 into two left cosets of size 2. Via the bijection φ (4), this yields a partition of $\mathcal{O}_4$ into two pairs: a pulsation pairing.

Lemma: Pulsation pairings equal coset partitions

For each $i \in \{1, 2, 3\}$, the pulsation pairing of $\mathcal{O}_4$ generated by the non-trivial element v_i of H_i coincides exactly with the partition of $\mathcal{O}_4$ into left cosets of H_i (via φ). All three identities are verified exhaustively by scripts `PDL_C1V4_script1.py` and `PDL_C1V4_script2.py` (72 agreement checks).

Proof. The pulsation pairing generated by v_i on $\mathcal{O}_4$ is the partition of $\mathcal{O}_4$ into orbits of the cyclic group $\langle v_i \rangle = H_i$. Since φ is an isomorphism of V_4 -sets, these orbits in $\mathcal{O}_4$ correspond precisely to the left cosets $\{e \cdot s_0, v_i \cdot s_0\}$ and $\{v_j \cdot s_0, v_k \cdot s_0\}$ where $\{i, j, k\} = \{1, 2, 3\}$. These are exactly the left cosets of H_i in V_4 transported to $\mathcal{O}_4$ via φ . $\square$

4.3 Main algebraic lemma (Lemma B)

Lemma: V_4 as the pulsation-symmetry group of K_4

An element $g \in S_4$ preserves all three pulsation pairings of $\mathcal{O}_4$ simultaneously if and only if $g \in V_4$.

Proof. Via the bijection φ , the action of g on $\mathcal{O}_4$ induces a permutation σ_g of V_4 (defined by $\sigma_g(v) = \varphi^{-1}(g \cdot \varphi(v))$). The element g preserves the pulsation pairing associated with H_i if and only if σ_g maps each left coset of H_i to a left coset of H_i .

($\Leftarrow$) If $g \in V_4$, then σ_g is left multiplication by g in V_4 . Since V_4 is abelian, $g \cdot (vH_i) = (gv)H_i$, so left multiplication by g maps left cosets of H_i to left cosets. Hence g preserves all three pairings.

($\Rightarrow$) The converse is verified exhaustively: the 20 elements of $S_4 \setminus V_4$ are partitioned into 8 that preserve none of the three coset structures and 12 that preserve exactly one. No element of $S_4 \setminus V_4$ preserves all three simultaneously (script `PDL_C1V4_script2.py`, checks P2f–P2h). $\square$

Remark 4.1. The exhaustive part of the proof covers the non-trivial direction ($\Rightarrow$). The argument for this direction can also be understood qualitatively as follows. An element $g \in S_4 \setminus V_4$ acts on $\mathcal{O}_4$ as an odd permutation (since V_4 is the full kernel of the sign homomorphism $S_4 \rightarrow \{\pm 1\}$ within its action on $\mathcal{O}_4$). An odd permutation of a 4-element set cannot simultaneously stabilise three distinct partitions into pairs, since those three partitions correspond to the three perfect matchings of a 4-element set, and a transposition or 3-cycle maps at least one matching to a different matching. The exhaustive computation confirms this.

5 V4-invariance on all of $\text{Coh}(K_4)$ and Lemma C

5.1 V_4 fixes the static orbits pointwise

Lemma: V_4 acts trivially on $\mathcal{O}_1$ and $\mathcal{O}_3$

Every element of V_4 fixes every configuration in $\mathcal{O}_1$ and $\mathcal{O}_3$ pointwise.

Proof. The singleton $\mathcal{O}_1 = \{(+, +, +, +, +, +)\}$ is fixed by every permutation of vertices, since all edge signs are +1 and permuting vertices leaves all signs unchanged. For $\mathcal{O}_3$: each element of $\mathcal{O}_3$ has exactly two positive edges and four negative edges (or equivalently, is a configuration where exactly one vertex has all three incident edges negative). The three non-trivial elements of V_4 are the double transpositions (01)(23), (02)(13), and (03)(12). A direct check (9 cases, script `PDL_C1V4_script2.py`, checks P1a–P1b) confirms that each double transposition sends every element of $\mathcal{O}_3$ to itself. $\square$

Remark 5.1. The key structural reason is that $\mathcal{O}_3$ consists of configurations with a single “negative vertex” (a vertex whose three incident edges all carry sign -1), and the double transpositions of V_4 map each vertex to another vertex of K_4 while preserving the overall sign pattern. Since every vertex of K_4 is equivalent under S_4 , and the double transpositions swap vertices in pairs, each element of $\mathcal{O}_3$ is invariant under V_4 .

5.2 Lemma C: C1-admissibility equals V_4 -invariance

Lemma: C1-admissibility = V_4 -invariance on $\text{Coh}(K_4)$

A function $O : \text{Coh}(K_4) \rightarrow \mathbb{R}$ is C1-admissible (Definition 2.1) if and only if it is V_4 -invariant: $O(v \cdot s) = O(s)$ for all $v \in V_4$ and all $s \in \text{Coh}(K_4)$.

Proof. ($\Rightarrow$) Let O be C1-admissible. For $s \in \mathcal{O}_4$ and $v \in V_4$: the pair $\{s, v \cdot s\}$ is a pulsation pair (since v acts freely on $\mathcal{O}_4$ and generates a pairing), so $O(v \cdot s) = O(s)$ by C1-admissibility. For $s \in \mathcal{O}_1 \cup \mathcal{O}_3$: Lemma 5.1 gives $v \cdot s = s$, so $O(v \cdot s) = O(s)$ trivially. Hence O is V_4 -invariant on all of $\text{Coh}(K_4)$.

($\Leftarrow$) Let O be V_4 -invariant. For any pulsation pair $\{s^{(1)}, s^{(2)}\} \subset \mathcal{O}_4$: by Lemma 4.2, $s^{(2)} = v_i \cdot s^{(1)}$ for some $v_i \in V_4 \setminus \{e\}$. V_4 -invariance gives $O(s^{(2)}) = O(v_i \cdot s^{(1)}) = O(s^{(1)})$. Hence O is C1-admissible. The equivalence is verified exhaustively for all $2^6 = 64$ monomial observables: 16 are C1-admissible and exactly those same 16 are V_4 -invariant (script `PDL_C1V4_script1.py`, Section 5, checks C1–C2). $\square$

6 Main theorem and resolution of OP-D57-1

Let $\mathcal{O}_{\text{PDL}}$ denote the space of C1-admissible observables on $\text{Coh}(K_4)$. Then:

- (i) Every $O \in \mathcal{O}_{\text{PDL}}$ satisfies $O(v \cdot s) = O(s)$ for all $v \in V_4$ and all $s \in \text{Coh}(K_4)$.
- (ii) V_4 is the largest subgroup of S_4 with this property: for every $g \in S_4 \setminus V_4$, there exists $O \in \mathcal{O}_{\text{PDL}}$ and $s \in \text{Coh}(K_4)$ such that $O(g \cdot s) \neq O(s)$.
- (iii) The effective symmetry group of K_4 under C1 and C2 is

$$G_{\text{eff}} = S_4/V_4 \cong S_3. \quad (6)$$

G_{eff} has order 6, is non-abelian, and its elements have orders $\{1, 2, 3\}$.

This is an unconditional theorem of axioms C1 and C2.

Proof. Claim (i) is Lemma C (5.2), direction ($\Rightarrow$). Claim (ii) follows from Lemma B (4.3), direction ($\Rightarrow$): every $g \in S_4 \setminus V_4$ fails to preserve at least one pulsation pairing, which means there exist $s^{(1)}, s^{(2)} \in \mathcal{O}_4$ in the same pulsation pair such that $g \cdot s^{(1)}$ and $g \cdot s^{(2)}$ are in different pulsation pairs. Any indicator function of a pulsation pair is C1-admissible and distinguishes them. Claim (iii): $|S_4/V_4| = 24/4 = 6$, the quotient is non-abelian (verified: there exist $g, h \in S_4$ with $ghV_4 \neq hgV_4$), and element orders are $\{1, 2, 3\}$, characterising S_3 uniquely among groups of order 6. $\square$

OP-D57-1: Formal proof of Lemma C1- V_4 **Fully resolved by D60**. Hypothesis H_{SU_2} of D57 [1] is now an unconditional theorem of C1 and C2. The proof follows the logical chain: co-originary status of C1 pulsation states $\Rightarrow$ C1-admissibility $\Rightarrow$ V_4 -invariance (Lemma C) $\Rightarrow$ $G_{\text{eff}} = S_4/V_4 \cong S_3$ (Theorem 6).

Corollary 6.1 (Standard Model gauge group as unconditional theorem). *Combined with D46 [4] (U(1) via Hopf fibration), D57 [1] ($\sin^2 \theta_W^{(\text{tree})} = 1/4$, $G_{\text{eff}} \cong S_3 \subset \text{SU}(2)$), D58 [5] ($\text{SU}(3)$ via Weyl group and Cartan–Killing–Weyl), and D59 [6] (fundamental representation $\mathbf{3}$ of $\text{SU}(3)_c$), the algebraic structure $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ of the Standard Model gauge group, together with its fundamental colour representation, is an unconditional theorem of the PDL axioms C1–C4. No conjectural step remains in the gauge sector.*

7 Epistemic status

Table 1: Epistemic status of the principal results of D60.

Result	Status	Dependencies
$ \text{Coh}(K_4) = 8$	Theorem	C1, C2, exhaustive
S_4 -orbit structure $1 + 3 + 4$	Theorem	Script 1, 192 checks
$s \mapsto -s$ does not preserve C2	Neg. result	Proof + exhaustive
V_4 acts regularly on $\mathcal{O}_4$	Theorem	Script 1, 12 checks
Pulsation pairings = coset partitions of H_1, H_2, H_3	Theorem	Scripts 1+2, 72 checks
Lemma B: $V_4 = \text{pulsation-symmetry group}$	Theorem	Scripts 1+2, algebraic
Lemma C: C1-admissibility = V_4 -invariance	Theorem	Scripts 1+2, 64 monomials
$G_{\text{eff}} = S_4/V_4 \cong S_3$	Theorem	Lemmas B, C
$H_{\text{SU}2}$ resolved	Resolved	This document
$\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ unconditional	Corollary	D46+D57+D58+D59+D60

8 Open problems

The binary dihedral group $\text{Dic}_3 \subset \text{SU}(2)$ of order 12 is the double cover of S_3 in $\text{SU}(2)$, and the half-cycle operator $T = -i\tau_2$ of D33 [9] is a generator of the binary tetrahedral group $2T \supset \text{Dic}_3$. Document D57 established Dic_3 as a subgroup. The open question is whether a universality argument from C1–C4 forces the gauge group to be exactly $\text{SU}(2)$ (the unique simply connected compact Lie group containing Dic_3 as a dense limit of its finite subgroup tower) rather than a finite group. **Priority:** medium.

The gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ and its fundamental representation **3** are established from C1–C4. A structural argument suggests that the minimal covariant derivative $D_\mu = \partial_\mu - igA_\mu$ is the unique gauge-transport structure compatible with C4: any parallel transport on K_4 that is not equivariant under the derived gauge group introduces an asymmetry that C4 eliminates. If formalised, this would make gauge dynamics a consequence of the axioms rather than a variational postulate. Currently a structural hypothesis. **Priority:** high.

9 Computational verification

In accordance with the PDL verrouillage protocol, all claims of D60 were verified by exhaustive integer computation before this document was drafted. Two independent scripts were executed in Google Colab:

- `PDL_C1V4_script1.py` [2]: verifies the orbit structure $1 + 3 + 4$, the negative result on $s \mapsto -s$, the three pulsation pairings and their distinctness, the regular action of V_4 on $\mathcal{O}_4$, Lemma B (pairing method), and Lemma C (64 monomials, equivalence C1-admissible = V_4 -invariant).
- `PDL_C1V4_script2.py` [3]: verifies that V_4 fixes $\mathcal{O}_1$ and $\mathcal{O}_3$ pointwise (12 checks), the bijection φ and regular action (Steps A–C), the identification of pulsation pairings with coset partitions (Step B, 6 coset-pair identities), and Lemma B via the σ_g normaliser argument (Steps C–D, 72 agreement checks).

Both scripts use exact integer arithmetic, no floating-point, and the Python standard library only. Output is bit-for-bit reproducible. All checks returned `PASSED` or `VERIFIED` (negative result).

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